

A Tale of Three Intersecting Lines

CHAPTER 07

Learning Outcomes...

- Triangle and its Terms
- Construction of Equilateral Triangles
- Forming Triangles Using Circles
- Triangle Inequality
- Understanding Triangle Existence Through Intersecting Circles
- Construction of Triangles When Some Sides and Angles are Given
- Angle Restriction for Triangle Construction
- Angle Sum Property and Exterior Angles of a Triangle
- Construction Related to Altitudes of Triangles
- Types of Triangles

Triangle and Its Terms

Triangle

A triangle is a simple closed figure formed by three straight lines or line segments.

Terms of a Triangle

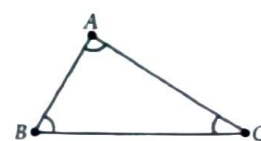
There are three terms of a triangle :

Side : A side of a triangle is one of the three straight line segments that forms the boundary of the triangle.

Vertex : A vertex of a triangle is a corner point where two sides of a triangle meet.

Angle : An angle of a triangle is the space formed between two sides that meet at a vertex.

For the given triangle ABC , $\overline{AB}$, $\overline{BC}$ and $\overline{CA}$ are sides; $\angle ABC$, $\angle BAC$, $\angle ACB$ are three angles and A , B , C are three vertices. It is denoted as $\triangle ABC$.



Construction of Equilateral Triangles

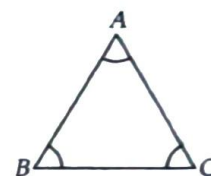
Equilateral Triangle

An equilateral triangle is a type of triangle whose all three sides are of equal length and all three angles are of equal measure. The equilateral triangles are the most symmetric ones.

For example ; In $\triangle ABC$, $AB = BC = CA$ and $\angle A = \angle B = \angle C$.

Construction of an Equilateral Triangle using Ruler

To construct an equilateral triangle having equal measure of side (say 3 cm) with just a ruler and pencil is a fundamental skill in geometry. Following are the given steps of constructing an equilateral triangle :



Step I : Draw the base $AB = 3$ cm using a ruler.

Step II : Measure 3 cm using the ruler and mark a point C from point A such that $AC = 3$ cm.

Step III : Join BC and check if $BC = 3$ cm. Adjust the position of point C until you get the exact measurement of $BC = 3$ cm.

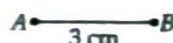
Step IV : When $BC = 3$ cm, then you get the required equilateral triangle ABC , with $AB = BC = CA = 3$ cm.

Limitation of Using a Ruler for Triangle Construction : Although it is possible to construct an equilateral triangle with a ruler, the process is inefficient because it depends on trial and error to find the third point C such that all sides are equal. To avoid this inefficiency, a compass is used.

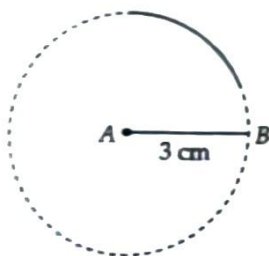
Construction of an Equilateral Triangle using Compass

To construct an equilateral triangle having equal measure of sides (say 3 cm) with the help of compass, the following steps need to be followed:

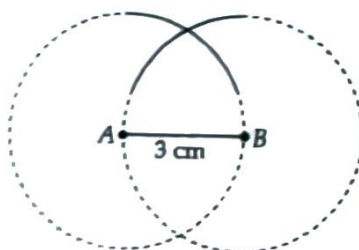
Step I : Start by constructing base, $AB = 3$ cm.



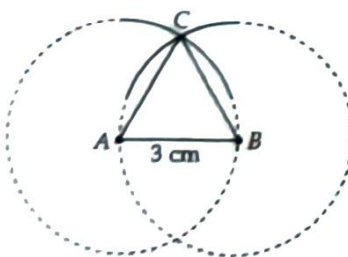
Step II : With centre as A and radius 3 cm, draw a sufficiently long arc (or circle) using compass.



Step III : With centre as B and radius 3 cm, draw another arc (or circle) of radius 3 cm.



Step IV : Make point of intersection as C . Join AC and BC .



Thus, $\triangle ABC$ is the required equilateral triangle.

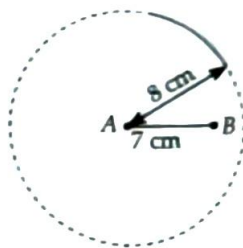
Construction of Triangle When its Sides are Given

Triangle can be constructed using compass when its three unequal sides (say $AB = 7$ cm, $BC = 9$ cm, $AC = 8$ cm) are given. Follow the given steps :

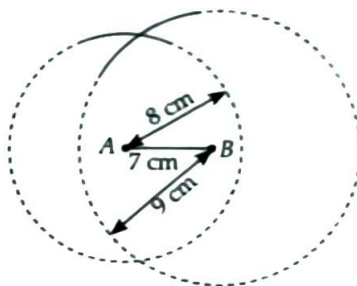
Step I : Draw the base $AB = 7$ cm using a ruler.



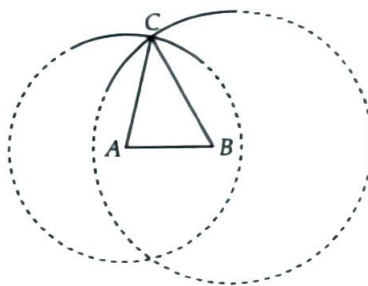
Step II : With A as center and compass width as 8 cm, draw an arc (or circle).



Step III : With B as center and compass width as 9 cm, draw an arc (or circle) intersecting the previous drawn arc.



Step IV : Mark point of intersection as C . Also, join AC and AB .



Thus, $\triangle ABC$ is the required triangle.

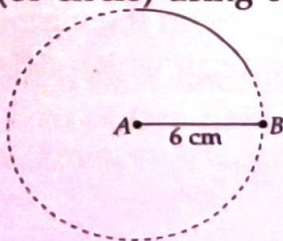
ILLUSTRATIONS

1 Construct an equilateral triangle with side length 6 cm using ruler and compass.

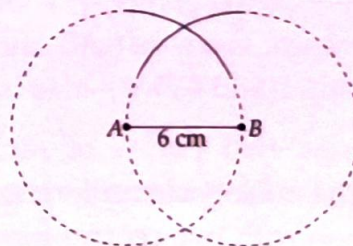
Soln.: Steps of construction :

Step I : Draw the base $AB = 6$ cm.

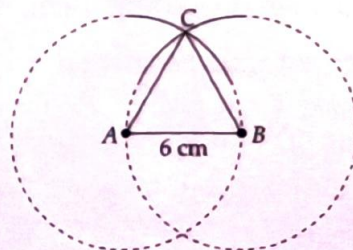
Step II : With centre as A and radius 6 cm, draw an arc (or circle) using compass.



Step III : Without changing the compass width, place the pointer at point B . Draw another arc (or circle) that intersects the previous drawn arc.



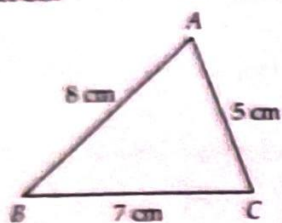
Step IV : Mark the point of intersection of the two arcs as point C and join points A to C and B to C using a ruler.



Thus, $\triangle ABC$ is the required equilateral triangle.

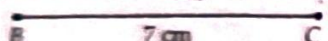
2 Construct a triangle ABC , given that $AB = 8$ cm, $BC = 7$ cm and $AC = 5$ cm.

Soln.: **Step I :** Draw a rough sketch with the given measures.

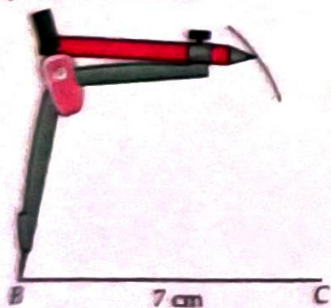


Rough Figure

Step II : Draw a line segment $BC = 7$ cm.



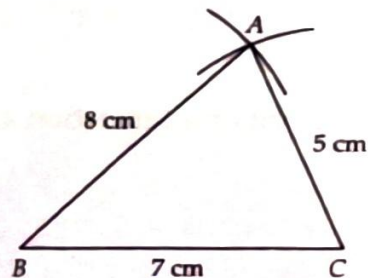
Step III : With B as centre, and compass width set to 8 cm, draw an arc.



Step IV : With C as centre and compass width set to 5 cm, draw another arc, that cut the previous arc drawn in Step III.



Step V : Mark the intersection point of both the arcs as A and join AB and AC .

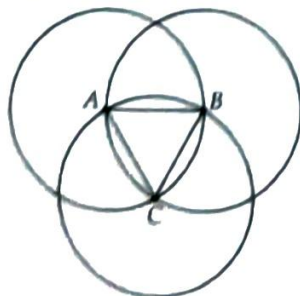
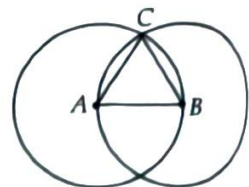
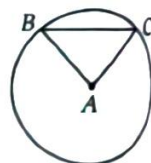


Therefore, $\triangle ABC$ is the required triangle.

Forming Triangles Using Circles

Circles can help us easily construct triangles with equal sides called equilateral triangles or those having two sides equal, called isosceles triangles.

- When we join the centre of a circle A , to two points B and C on circumference, such that AB and AC are radii of the circle. Then after joining B and C , we get an isosceles triangle in which $AB = AC$.
- If we draw two circles of same radius such that their center lies on the others circumference, then on joining their point of intersection with two centres forms an equilateral triangle in which $AB = BC = AC$.
- If three circles of equal radius are drawn so that each centre lies on the circumference of the other two, then on joining them, they form an equilateral triangle in which $AB = AC = BC$.



Triangle Inequality

A triangle is possible for some set of lengths and not possible for others.

For a triangle to exist, each length is smaller than the sum of the other two.

Mathematically, In $\triangle ABC$, let $BC = a$, $AC = b$ and $AB = c$. Then, $a < b + c$; $b < c + a$; $c < a + b$.

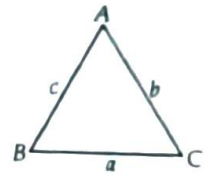
This is also known as triangle inequality.

If any one of the inequality fails, then the triangle is not possible.

For example ; a triangle with sides 5 cm, 2 cm and 2 cm is not possible as $5 \nless 2 + 2$.

In a triangle, the length of any side is always greater than the absolute difference of the other two sides. If the sides of a triangle are denoted by a , b , and c , then the following inequalities hold,

$$|a - b| < c; |a - c| < b; |b - c| < a$$



ILLUSTRATIONS

- 3** Is it possible to draw a triangle, the lengths of whose sides are 3 cm, 4 cm and 5 cm?

Soln.: Consider the numbers 3, 4, 5.

$$3 < 4 + 5 = 9; 4 < 3 + 5 = 8; 5 < 4 + 3 = 7$$

It is clear that the sum of any two of these numbers is greater than the third side.

Hence, it is possible to draw a triangle whose sides are 3 cm, 4 cm and 5 cm.

- 4** Is it possible to draw a triangle with sides 5 cm, 7 cm and 12 cm?

Soln.: Consider the sides as 5, 7, 12.

Clearly, $5 + 7 = 12$.

Thus, the sum of two sides is not greater than the third side.

Hence, it is not possible to draw a triangle with sides 5 cm, 7 cm and 12 cm.

- 5** Three friends are standing at point A, B and C in a park such that ABC forms a triangle if $AB = 8$ m and $BC = 5$ m then what could be the length of AC?

Soln.: We are given that ABC forms a triangle and $AB = 8$ m, $BC = 5$ m.

$\therefore$ By triangle's inequality

$$AC < AB + BC$$

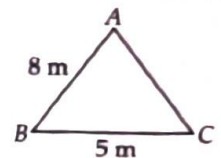
$$\Rightarrow AC < 8 + 5 = 13 \text{ m}$$

$\therefore$ Length of AC must be less than 13 m.

Also, $AC + BC > AB$

$$\Rightarrow AC + 5 > 8, \text{ which will always hold when } AC > 3.$$

$\therefore$ AC must be greater than 3 m and less than 13 m.

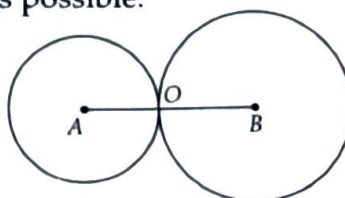


Understanding Triangle Existence Through Intersecting Circles

When constructing a triangle, we take the longest side as base and then draw arcs (or circles) from each end of base with radii equal to the other two side lengths. There are three possible cases for constructing circle to determine whether a triangle is possible or not.

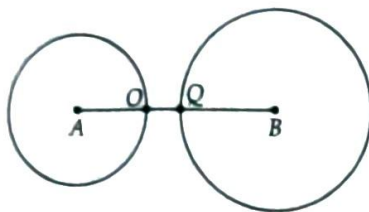
Case I : Circles touch each other at a point.

In this case, the sum of two radii = AB i.e., $AO + OB$ or sum of the two smaller lengths = longest length of triangle. Thus, no triangle is possible.



Case II : Circles do not intersect internally.

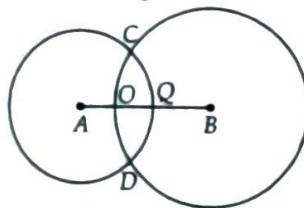
In this case, sum of two radii is less than AB or sum of the two smaller lengths $<$ longest length of triangle (i.e., $AO + BQ < AB$)



Thus, no triangle is possible.

Case III : Circles intersect each other.

In this case, sum of the two radii is greater than AB or sum of the two smaller lengths $>$ longest length (i.e., $AO + BQ > AB$) [$\because OQ$ is added twice]



In such case, we have 2 point of intersections as C and D that gives us two triangles, one above the line AB and one below the line AB .

Here, the rule of triangle inequality is satisfied as sum of two smaller lengths $>$ longest length. Hence, only in such case triangle exists.

ILLUSTRATIONS

6 Is it possible to draw a triangle with two sides equal and one different? Give example and draw the triangle.

Soln.: Let the sides of triangle be $AB = 5$ cm, $BC = 3$ cm and $AC = 3$ cm.

Since, $5 < 3 + 3 = 6$; $3 < 5 + 3 = 8$

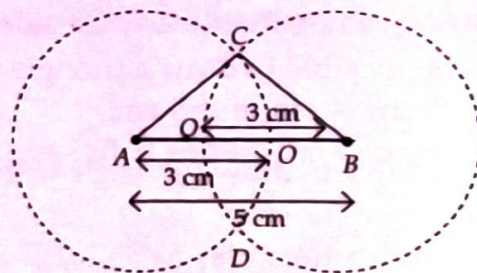
$\therefore$ Triangle exists and the triangle with such measure can be drawn by following the given steps :

Step I : Draw a line AB (longest side) = 5 cm.

Step II : Take A and B as centre and draw two circles of radius 3 cm each, intersecting each other at two points, C and D .

Step III : Join either AC and BC or AD and

BD . Thus, we get the required triangle.



7 Two circles have radii 6 cm and 4 cm. The distance between this centres is 12 cm. Will the circles touch, intersect at two points or do not intersect? Is the triangle possible?

Soln.: Since, $4 + 6 = 10 < 12$, so the sum of the radii less than the longest length.

Thus, circles do not intersect at any point and hence no triangle is possible.

Construction of Triangles When Some Sides and Angles are Given

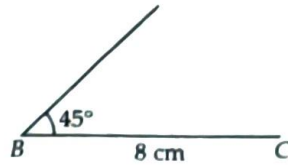
Two Sides and the Included Angle

In this case, two sides and the angle between the both sides, i.e., the included angle is given. To construct a triangle ABC when two sides ($AB = 6$ cm, $BC = 8$ cm) and the angle between them, $\angle ABC = 45^\circ$ (called the included angle) are given, follow the given steps :

Step I : Draw the base $BC = 8$ cm using a ruler.

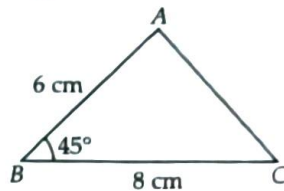


Step II : Construct $\angle B = 45^\circ$ by drawing the other arm of the angle.



Step III : Mark the point A on the other arm such that $AB = 6$ cm using a compass.

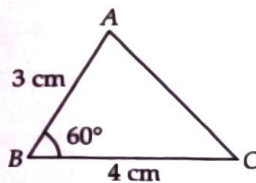
Step IV : Join AC to get the required triangle.



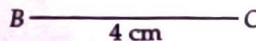
ILLUSTRATIONS

8 Draw a $\triangle ABC$ with $AB = 3$ cm, $BC = 4$ cm and $\angle ABC = 60^\circ$.

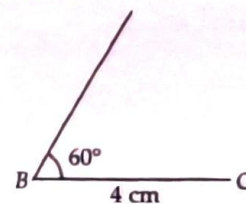
Soln.: Step I : Draw a rough sketch with the given measures.



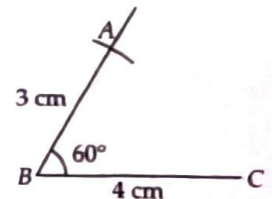
Step II : Draw a line segment BC of length 4 cm using a ruler.



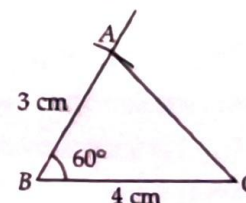
Step III : Construct $\angle B = 60^\circ$ using protractor by drawing the other arm of the triangle.



Step IV : Using compass and taking B as centre, mark an arc of length 3 cm on B, the point of intersection will be point A.



Step V : Join AC to get the required triangle ABC.



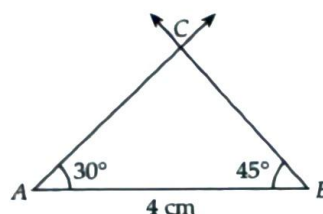
Two Angles and the Included Side

In this case, two angles and the side that is a part of both the angles *i.e.*, the included side is given. To construct a triangle ABC when two angles ($\angle A = 30^\circ$ and $\angle B = 45^\circ$) and the side between them, $AB = 4$ cm (called the included side) are given, follow the given steps:

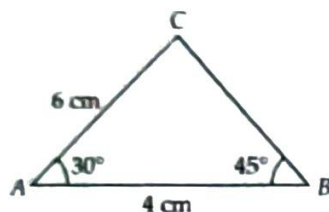
Step I : Draw the base $AB = 4$ cm using a ruler.



Step II : Construct $\angle A = 30^\circ$ and $\angle B = 45^\circ$ by drawing the other arm of the angle.



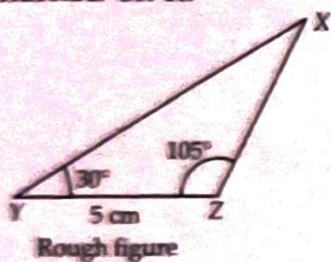
Step III : Let C be a point of intersection of both the rays of the angle. Join AC and BC to get the required triangle ABC .



ILLUSTRATIONS

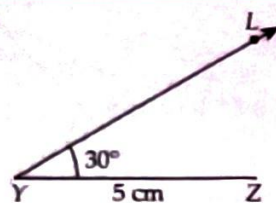
9 Construct a triangle XYZ with $YZ = 5$ cm, $\angle Y = 30^\circ$ and $\angle Z = 105^\circ$.

Soln.: **Step I :** Draw a rough sketch with measures marked on it.



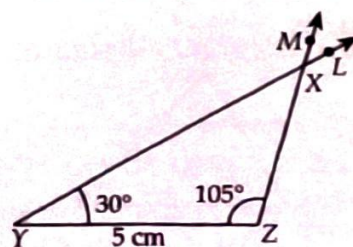
Step II : Draw a line segment YZ of length 5 cm.

Step III : At Y , draw a ray YL making an angle of 30° with YZ .



Step IV : At Z , draw a ray ZM making an angle of 105° with YZ and let X is the point of intersection of both the rays.

Then, ΔXYZ is the required triangle.



Angle Restriction for Triangle Construction

In previous section, we see that two angles and the included side forms a triangle but not always. It is possible that the two rays made do not meet at any point. There are certain angle combination where the two sides drawn from the base will never meet to form a triangle.

If both given angles are 90° or more than 90° , they form the sides too far from each other such that the sides never meet.

For example ; If two angles are 100° each, then they form the lines that never meet.



Even if one angle is acute and the other angle is obtuse (very large) then also the formed sides will either be parallel or move away from each other, so no triangle is possible.

For example ; Let two angles be 40° and 150° , then the formed sides will not meet.



Note : Length of base does not affect the existence of triangle, it entirely depends on the given angles.

Angle Sum Property of a Triangle

To determine whether a triangle is possible from the given measures of angles, we use the angle sum property of a triangle which states that "The sum of all the three angles of a triangle is always 180° "

To justify this property, we proceed as follows:

Consider a triangle ABC . From A , draw a line DE parallel to BC .

Since, $DE \parallel BC$

$$\angle DAB = \angle ABC$$

[Alternate angles]

$$\angle EAC = \angle ACB$$

[Alternate angles]

$$\text{Now, } \angle DAC + \angle CAE = 180^\circ$$

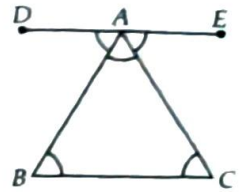
[Linear pair]

$$\Rightarrow \angle DAB + \angle BAC + \angle CAE = 180^\circ$$

$$\Rightarrow \angle ABC + \angle BAC + \angle ACB = 180^\circ$$

This property allows us to find the missing angle of any triangle without constructing it. If the sum of any two given angles is less than 180° , a triangle can exist, but if the sum is equal to or greater than 180° , no triangle is possible.

For example ; If measure of two angles of a triangle are 60° and 70° then $60^\circ + 70^\circ < 180^\circ$, so, triangle is possible and measure of the third angle will be $180^\circ - 60^\circ - 70^\circ = 50^\circ$.



ILLUSTRATIONS

10 In triangle ABC , $\angle A = 2\angle B$ and $\angle C = 3\angle B$ then find the measure of all the three angles.

Soln.: For any triangle, the sum of all angles is equal to 180° , so for $\triangle ABC$, we have

$$\angle A + \angle B + \angle C = 180^\circ$$

$$\Rightarrow 2\angle B + \angle B + 3\angle B = 180^\circ$$

$$\Rightarrow 6\angle B = 180^\circ$$

$$\Rightarrow \angle B = \frac{180^\circ}{6} = 30^\circ$$

$$\therefore \angle A = 2 \times 30^\circ = 60^\circ \text{ and } \angle C = 3 \times 30^\circ = 90^\circ$$

11 Find the measure of third angle for the given two angles of a triangle (if exists).

(a) 60° , 50°

(b) 70° , 110°

Determine whether the triangle can be formed in each case or not.

Soln.: (a) Sum of all angles in a triangle is 180° .

Let measure of third angle be x° .

$$\therefore 60^\circ + 50^\circ + x^\circ = 180^\circ$$

$$\Rightarrow x^\circ = 180^\circ - 60^\circ - 50^\circ$$

$$\Rightarrow x^\circ = 70^\circ$$

$\therefore$ Measure of third angle is 70° .

In such case, triangle is possible.

(b) Let measure of third angle be x° .

$$\therefore 70^\circ + 110^\circ + x^\circ = 180^\circ$$

$$\Rightarrow x^\circ = 180^\circ - 70^\circ - 110^\circ$$

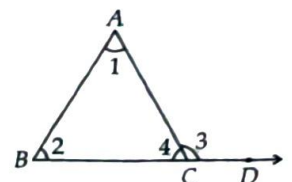
$$\Rightarrow x^\circ = 0^\circ$$

In such case, triangle is not possible.

Exterior Angles of a Triangle

An exterior angle of a triangle is formed when we produce any of the side of triangle in either direction. The exterior angle property states that an exterior angle of a triangle is equal to the sum of its interior opposite angles.

In the given $\triangle ABC$, side BC is produced to D . $\angle ACD$ is the exterior angle of $\triangle ABC$ such that $\angle 3 = \angle 1 + \angle 2$.



The exterior angle property can be proved by using angle sum property of a triangle.

In $\triangle ABC$, $\angle 1 + \angle 2 + \angle 4 = 180^\circ$ [Angle sum property]

Also, $\angle 3 + \angle 4 = 180^\circ$ [Linear pair]

$$\Rightarrow \angle 1 + \angle 2 + \angle 4 = \angle 3 + \angle 4 \Rightarrow \angle 1 + \angle 2 = \angle 3$$

Hence, proved.

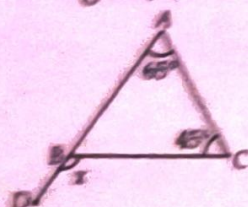
- Exterior angle and its corresponding interior angle form a linear pair.
- When an exterior angle is acute angle, then both interior opposite angles are also acute angles.

COMPETITION WINDOW

- When an exterior angle is obtuse angle, then both interior angles may be acute or one of them is obtuse.

ILLUSTRATIONS

- 12 In the given figure, find the value of x .

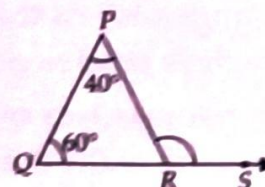


Soln.: In $\triangle ABC$, $\angle CBD$ is the exterior angle of triangle.

$$\therefore \angle CBD = \angle BCA + \angle BAC$$

$$\Rightarrow \angle x = 45^\circ + 65^\circ = 110^\circ$$

- 13 In $\triangle PQR$, find the exterior angle, $\angle PRS$.



Soln.: The exterior angle of a triangle is equal to the sum of its interior opposite angles.

$$\therefore \angle PRS = \angle QPR + \angle PQR$$

$$\Rightarrow \angle PRS = 40^\circ + 60^\circ$$

$$\Rightarrow \angle PRS = 100^\circ$$

Altitudes of a Triangle

In a triangle, altitude is the perpendicular distance from any vertex to its opposite side. This distance is measured along perpendicular line making it "height" of the triangle from that base. In $\triangle ABC$, AD is the altitude as $AD \perp BC$.

The altitude of a triangle has some properties :

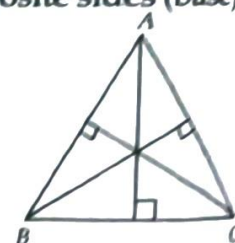
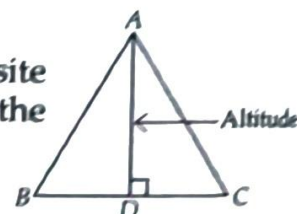
- The altitudes can be drawn from any of the three vertices to their respective opposite sides (base).
- The choice of base changes according to vertex and which altitude is considered.

If BC is the base, the altitude is drawn from vertex A .

If AC is the base, the altitude is drawn from vertex B .

If AB is the base, the altitude is drawn from vertex C .

Note : Sometimes, the height can even fall outside the triangle, as in case of obtuse angled triangle.



Constructions Related to Altitudes of Triangles

Using Paper Folding

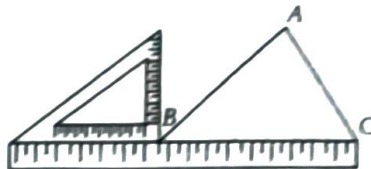
Take a paper triangle and choose one of its side as the base. Fold the paper in such a way that a straight line is formed from the top vertex to the base.

The crease formed is the altitude, which is perpendicular to the base.

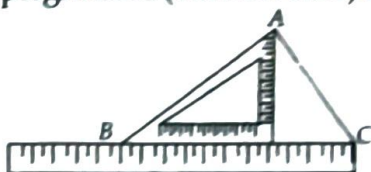
Using Set Square and Ruler

Construct a $\triangle ABC$ and choose one of its side as a base. Let BC be the base. Now, constructing an altitude using just a ruler may not give accurate line that meets base at 90° . So, we use a set square to draw the altitude to get perfect 90° line by following the given steps :

Step I : Place ruler along the base BC and put the set square on the ruler such that one of its side is at right angle with ruler as shown.



Step II : Slide the set square till the upright side (vertical side) of the set square touches the vertex A .



Step III : Draw the line along set square from vertex A to base BC . We get the required altitude which is perpendicular to the base.

Note : In right angled triangle, the side which is perpendicular to the base is itself the altitude to that base.

The three altitudes of a triangle meet at a single point called the orthocentre of the triangle.

COMPETITION WINDOW



- » In acute angled triangle, orthocentre lies inside the triangle.
- » In right angled triangle, it lies at the vertex of the right angle.
- » In obtuse angled triangle, it lies outside the triangle.

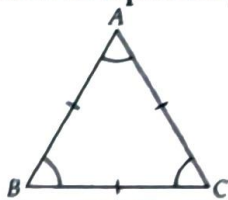
Types of Triangles

The triangles can be classified on the basis of

- (i) Sides (ii) Angles

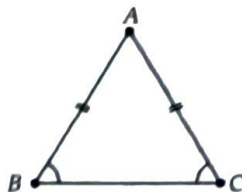
(i) On the basis of sides, the triangles are classified as :

Equilateral triangle : A triangle with all sides of equal length and all angles of same measure as 60° .



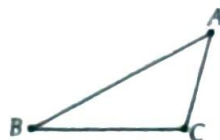
In $\triangle ABC$, $AB = BC = CA$ and $\angle ABC = \angle BCA = \angle BAC = 60^\circ$

Isosceles triangle : A triangle with two sides are of equal length and two angles opposite to the two equal sides are of equal measure.



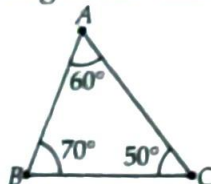
In $\triangle ABC$, $AB = AC$, $\angle ABC = \angle ACB$.

Scalene triangle : A triangle with all the three sides of different lengths and all angles are of different measure.

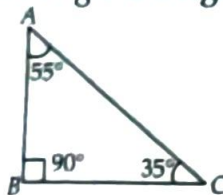


(ii) On the basis of angles, the triangles are classified as :

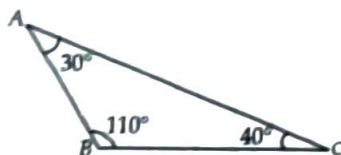
Acute angled triangle : All three interior angles are acute angles (less than 90°).



Right angled triangle : One of the interior angle is a right angle (equal to 90°).



Obtuse angled triangle : One of the interior angle is an obtuse angle (greater than 90°).



ILLUSTRATIONS

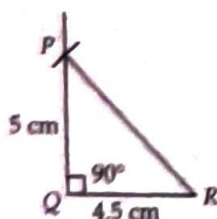
14 Construct a right-angled triangle PQR in which $\angle Q = 90^\circ$, $QR = 4.5$ cm and $PQ = 5$ cm. Draw the altitude from R to PQ .

Soln.: **Step I :** Draw $QR = 4.5$ cm

Step II : At Q , construct a right angle (90°) and mark $PQ = 5$ cm on perpendicular line.

Step III : Join PR .

$\therefore RQ$ is the required altitude from R to PQ as $\angle RQP = 90^\circ$.



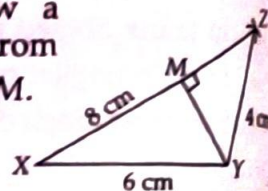
15 Construct an obtuse-angled triangle XYZ with $XY = 6$ cm, $YZ = 4$ cm and $ZX = 8$ cm. Draw the altitude from Y .

Soln.: **Step I :** Draw $XY = 6$ cm.

Step II : With X as centre, draw an arc of radius 8 cm and with Y as centre, draw an another arc of radius 4 cm intersecting at Z .

Step III : Join XZ and YZ .

Step IV : Now draw a perpendicular line on XZ from Y which intersect XZ at M .
 $\therefore YM$ is the required altitude from Y to XZ .



16 Construct an acute-angled triangle ABC with $AB = 8$ cm, $BC = 5$ cm and $AC = 9$ cm. Draw all three altitude.

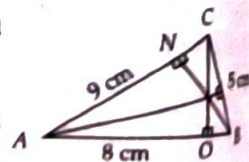
Soln.: **Step I :** Draw $AB = 8$ cm.

Step II : With A as centre, draw an arc of radius 9 cm and with B as centre, draw an another arc of radius 5 cm intersecting at C .

Step III : Join AC and BC .

Step IV : To draw altitudes,

- From A , draw a perpendicular to BC .
 - From B , draw a perpendicular to AC .
 - From C , draw a perpendicular to AB .
- $\therefore AM, BN$ and CO are required altitudes.



A TALE OF THREE INTERSECTING LINES

Triangle

- A triangle is the close figure formed by three straight lines.

Angle

The space formed between two sides that meet at a vertex.

Side

The straight line forming the boundary of the triangle.

Vertices

A corner point where two sides of a triangle meet.

Construction of a triangle is possible for three cases:

- Three sides are given.
- Two sides and the included angle is given.
- Two angles and the included side is given.

Triangle Inequality

The sum of the lengths of any two sides of a triangle must be greater than the length of the third side.

Angle Sum Property

The sum of all three interior angles of a triangle is equal to 180° .

Exterior Angle Property

An exterior angle of a triangle is equal to the sum of its two opposite interior angles.

Altitude

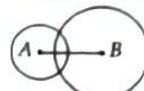
Altitude of a triangle is a perpendicular line drawn from a vertex of the triangle to its opposite side.

Constructing an altitude using paper folding and only a ruler is possible but inefficient than constructing an altitude using a set square along with a ruler.

3 possible cases for constructing triangles using circles



Case I: sum of two radii of the circle = Length of AB
 $\therefore$ No triangle is possible



Case II: sum of two radii of the circle > Length of AB
 Triangle exists



Case III: sum of two radii of the circle < Length of AB
 $\therefore$ No triangle is possible

Types of triangle

Classification Based on Sides

Equilateral Triangle

All three sides are equal in length, and all three angles measures 60° .

Isosceles Triangle

Two sides are equal in length, and the angles opposite to those sides are also equal.

Scalene Triangle

All three sides are of different lengths, and all three angles are of different measures.

Classification Based on Angles

Acute Triangle

All three interior angles are less than 90° .

Obtuse Triangle

One interior angle is greater than 90° .

Right Triangle

One interior angle is exactly 90° .

Construction of an Equilateral Triangle

- Constructing an equilateral triangle using only a ruler is possible but inefficient than constructing an equilateral triangle using a compasses and a ruler.

Solved Examples

1. Construct a $\triangle ABC$ in which $BC = 5.2$ cm, $AB = 4$ cm and $AC = 3.3$ cm.

Soln.: First we draw a rough sketch of $\triangle ABC$ and write down its dimensions as shown.



Steps of Construction :

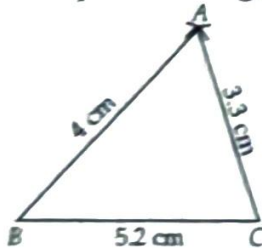
Step I : Draw a line segment $BC = 5.2$ cm.

Step II : With B as centre and radius 4 cm, draw an arc.

Step III : With C as centre and radius 3.3 cm, draw another arc, cutting the previous arc at A .

Step IV : Join AB and AC .

$\therefore \triangle ABC$ is the required triangle.



2. Construct a $\triangle ABC$ in which $AB = 5$ cm, $AC = 4.3$ cm and $\angle A = 60^\circ$.

Soln.: First we draw a rough sketch of $\triangle ABC$ and write down its dimensions as shown.



Steps of Construction :

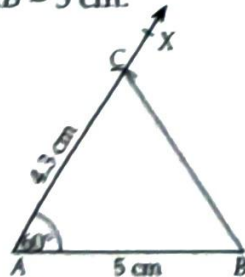
Step I : Draw a line segment $AB = 5$ cm.

Step II : Draw $\angle BAX = 60^\circ$.

Step III : With A as centre, draw an arc of radius 4.3 cm on AX and mark the point of intersection as C .

Step IV : Join BC .

$\therefore \triangle ABC$ is the required triangle.



3. Draw $\triangle PQR$ in which $QR = 6$ cm, $\angle Q = 35^\circ$ and $\angle R = 100^\circ$. Also, measure $\angle P$.

Soln.: First we draw a rough sketch of $\triangle PQR$ and write down its dimensions as shown.

Steps of Construction :

Step I : Draw a line segment $QR = 6$ cm.

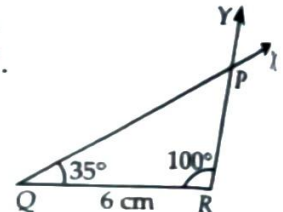
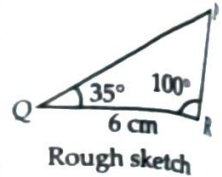
Step II : Draw $\angle RQX = 35^\circ$.

Step III : Draw $\angle QRY = 100^\circ$.

Step IV : Let the rays QX and RY intersect at P .

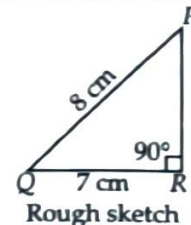
$\therefore \triangle PQR$ is the required triangle.

By measurement we find that $\angle P = 45^\circ$.



4. Construct a triangle PQR , right-angled at R in which $PQ = 8$ cm and $QR = 7$ cm.

Soln.: First we draw a rough sketch of $\triangle PQR$ and write down its dimensions as shown.



Steps of Construction :

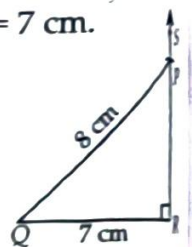
Step I : Draw a line-segment $QR = 7$ cm.

Step II : At R , draw $\angle QRS = 90^\circ$.

Step III : With Q as centre and radius equal to 8 cm, draw an arc cutting RS at P .

Step IV : Join QP .

$\therefore \triangle PQR$ is the required triangle.



5. In the given figure, P is a point on the side BC of $\triangle ABC$. Prove that $(AB + BC + AC) > 2AP$.



Soln.: We know that in a triangle, the sum of any two sides is greater than the third side.

In $\triangle ABP$; $AB + BP > AP$... (i)

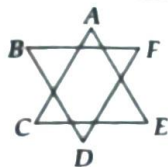
In $\triangle APC$, $PC + AC > AP$... (ii)

Now, adding (i) and (ii), we get

$AB + BP + PC + AC > AP + AP$

$\Rightarrow AB + BC + AC > 2AP$

6. The adjoining figure has been obtained by using two triangles. Prove that $\angle A + \angle B + \angle C + \angle D + \angle E + \angle F = 360^\circ$.



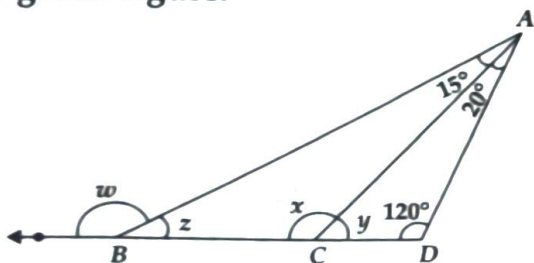
Soln.: We know that the sum of the angles of a triangle is 180° .

In $\triangle ACE$; $\angle A + \angle C + \angle E = 180^\circ$... (i)

In $\triangle BDF$; $\angle B + \angle D + \angle F = 180^\circ$... (ii)

On adding (i) and (ii), we get
 $\angle A + \angle B + \angle C + \angle D + \angle E + \angle F = 360^\circ$

7. Find the values of the unknown variables in the given figure.



Soln.: In $\triangle ACD$,
 $20^\circ + 120^\circ + y = 180^\circ$ (Angle sum property)

$\Rightarrow y = 180^\circ - 120^\circ - 20^\circ = 40^\circ$

Also, $x = 20^\circ + 120^\circ$ (Exterior angle property)
 $\Rightarrow x = 140^\circ$

In $\triangle ABC$,
 $15^\circ + x + z = 180^\circ$ (Angle sum property)

$\Rightarrow z = 180^\circ - x - 15^\circ = 180^\circ - 140^\circ - 15^\circ = 25^\circ$

Now, $w + z = 180^\circ$ (Linear pair)

$\Rightarrow w = 180^\circ - z = 180^\circ - 25^\circ = 155^\circ$

Hence, $x = 140^\circ$, $y = 40^\circ$, $z = 25^\circ$ and $w = 155^\circ$

8. Check, whether the given dimensions represent the sides of a triangle or not.

(i) 5 cm, 7 cm and 9 cm

(ii) 1.8 cm, 2.8 cm and 4.8 cm

Soln.: (i) $5 \text{ cm} + 7 \text{ cm} = 12 \text{ cm} > 9 \text{ cm}$

$5 \text{ cm} + 9 \text{ cm} = 14 \text{ cm} > 7 \text{ cm}$

$7 \text{ cm} + 9 \text{ cm} = 16 \text{ cm} > 5 \text{ cm}$

Hence, 5 cm, 7 cm and 9 cm represent the sides of a triangle.

(ii) $1.8 \text{ cm} + 2.8 \text{ cm} = 4.6 \text{ cm} < 4.8 \text{ cm}$

Hence, 1.8 cm, 2.8 cm and 4.8 cm do not represent the sides of triangle.

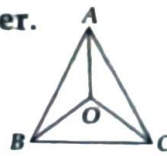
9. O is a point in the interior of $\triangle ABC$. State whether the following statements are true or false. Give reasons for your answer.

(i) $OA + OB > AB$

(ii) $OA + OC < AC$

(iii) $OB + OC < BC$

(iv) $AB + BC + AC = 2(OA + OB + OC)$



Soln.: (i) OA, OB and AB are the sides of $\triangle OAB$.
 $\therefore OA + OB > AB$ (Triangle inequality)

Hence, the given statement is true.

(ii) OA, OC and AC are the sides of $\triangle OAC$.

$\therefore OA + OC > AC$ (Triangle inequality)

Hence, the given statement is false.

(iii) OB, OC and BC are the sides of $\triangle OBC$.

$\therefore OB + OC > BC$ (Triangle inequality)

Hence, the given statement is false.

(iv) Adding the 3 statements proved above, we get

$OA + OB + OA + OC + OB + OC > AB + AC + BC$

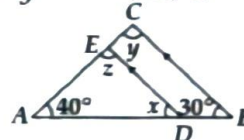
$\Rightarrow 2(OA + OB + OC) > AB + AC + BC$

or $AB + AC + BC < 2(OA + OB + OC)$

Hence, the given statement is false.

10. In figure, D and E are the points on sides AB and AC of $\triangle ABC$ such that $DE \parallel BC$. If $\angle B = 30^\circ$ and $\angle A = 40^\circ$, find :

(i) x (ii) y (iii) z



Soln.: (i) $\angle ADE = \angle ABC$
 (Corresponding angles as $DE \parallel BC$)

$\Rightarrow x = 30^\circ$ ($\because \angle ABC = 30^\circ$)

(ii) In $\triangle ABC$, we have

$\angle A + \angle B + \angle C = 180^\circ$ (Angle sum property)

$\Rightarrow 40^\circ + 30^\circ + y = 180^\circ$

$\Rightarrow 70^\circ + y = 180^\circ \Rightarrow y = 180^\circ - 70^\circ = 110^\circ$

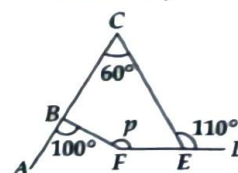
(iii) In $\triangle ADE$, we have,

$x + 40^\circ + z = 180^\circ$ (Angle sum property)

$\Rightarrow 30^\circ + 40^\circ + z = 180^\circ \Rightarrow 70^\circ + z = 180^\circ$

$\Rightarrow z = 180^\circ - 70^\circ = 110^\circ$

11. In the figure, ABC and FED are straight lines. Find the value of p.



Soln.: Join CF.

$$z = 180^\circ - 110^\circ = 70^\circ$$

[Linear pair]

$$y = 180^\circ - 100^\circ = 80^\circ$$

[Linear pair]

In $\triangle CFE$,

$$\angle CFE + \angle FCE + \angle FEC = 180^\circ$$

(Angle sum property) ... (i)

$$\text{In } \triangle BCF, \angle BCF + \angle BFC + \angle CBF = 180^\circ \quad \dots (ii)$$

(Angle sum property)

Adding (i) and (ii), we get

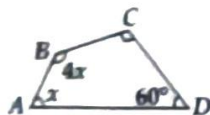
$$\angle CFE + \angle BFC + \angle FCE + \angle BCF + \angle CBF + \angle FEC = 180^\circ + 180^\circ$$

$$\Rightarrow \angle BFE + \angle BCE + \angle CBF + \angle FEC = 360^\circ$$

$$\Rightarrow p + 60^\circ + y + z = 360^\circ$$

$$\Rightarrow p = 360^\circ - 60^\circ - y - z \\ = 360^\circ - 60^\circ - 80^\circ - 70^\circ = 150^\circ$$

12. Find the value of x for the given figure.



Soln.: Join BD.

In $\triangle ABD$; $\angle ABD + \angle ADB$

$$+ \angle BAD = 180^\circ \quad \dots (i)$$

[Angle sum property]

In $\triangle BCD$; $\angle BCD + \angle CBD$

$$+ \angle BDC = 180^\circ \quad \dots (ii)$$

(Angle sum property)

Adding (i) and (ii), we get

$$\angle BAD + \angle ABD + \angle CBD + \angle BCD + \angle ADB + \angle BDC$$

$$= 180^\circ + 180^\circ$$

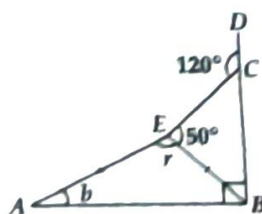
$$\Rightarrow \angle BAD + \angle ABC + \angle BCD + \angle ADC = 360^\circ$$

$$\Rightarrow x + 4x + 90^\circ + 60^\circ = 360^\circ$$

$$\Rightarrow 5x + 150^\circ = 360^\circ \Rightarrow 5x = 360^\circ - 150^\circ = 210^\circ$$

$$\Rightarrow x = 42^\circ$$

13. In the figure, BCD is a straight line, $\angle EAB = \angle EBA$ and $\angle ABC = 90^\circ$. Find the value of r .



$$\text{Soln.} \angle EBC + 50^\circ = 120^\circ$$

[Exterior angle property]

$$\Rightarrow \angle EBC = 70^\circ$$

$$\therefore \angle EBA = \angle ABC - \angle EBC = 90^\circ - 70^\circ = 20^\circ$$

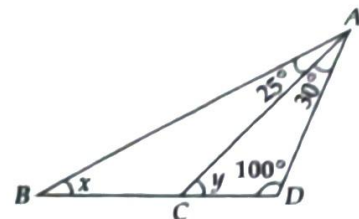
Given $\angle EAB = \angle EBA = 20^\circ$

In $\triangle ABE$,

$$r + 20^\circ + 20^\circ = 180^\circ \quad (\text{Angle sum property})$$

$$\Rightarrow r = 180^\circ - 40^\circ = 140^\circ$$

14. Find the values of the unknown variables in the figure given below.



Soln.: In $\triangle ACD$,

$$y + 100^\circ + 30^\circ = 180^\circ \quad (\text{Angle sum property})$$

$$\Rightarrow y + 130^\circ = 180^\circ \Rightarrow y = 180^\circ - 130^\circ = 50^\circ$$

For $\triangle ABC$, y is an exterior angle

$$\therefore x + 25^\circ = y \quad (\text{Exterior angle property})$$

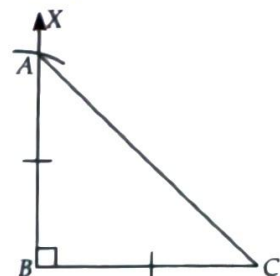
$$\Rightarrow x + 25^\circ = 50^\circ \Rightarrow x = 50^\circ - 25^\circ = 25^\circ$$

15. Construct an isosceles right angled triangle whose base measures 4 cm and draw its altitudes.

Soln.: Step I : Draw a base BC of length 4 cm.

Step II : Draw an angle of 90° on vertex B.

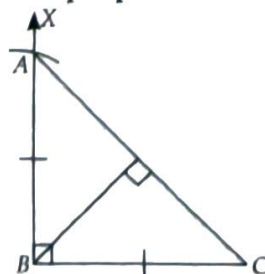
Step III : Mark an arc of radius 4 cm on BX such that it cuts BX at A.



Step IV : Join A and C.

Now, $\triangle ABC$ is a right angled isosceles triangle where $BC = AB = 4$ cm.

In a right angled triangle, two of its side which are perpendicular to each other are its altitudes. Now, for third altitude taking AC as base and B as vertex draw BD perpendicular to AC from B.



16. The lengths of two sides of a triangle are 12 cm and 15 cm. Between what two measures should the length of the third side fall?

Soln.: Lengths of two sides of a triangle are 12 cm and 15 cm.

In a triangle, the sum of the lengths of any two side is always greater than the third side.

Let the third side be x .

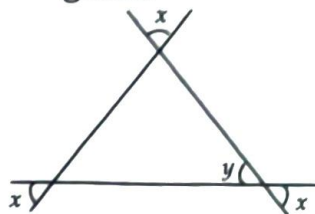
$$\therefore x + 12 > 15 \Rightarrow x > 3$$

Also, $x + 15 > 12 \Rightarrow x > -3$ but side length can never be -ve.

$$\text{and } 12 + 15 > x \Rightarrow 27 > x$$

Hence, third side can measure between 3 cm and 27 cm.

17. Find the value of the unknown x and y in the given diagram.



Soln.: $y = x$ (Vertically opposite angles)

$a = x$ (Vertically opposite angles)
 $b = x$ (Vertically opposite angles)

$$\text{Now, } a + b + y = 180^\circ$$

(Angle sum property)

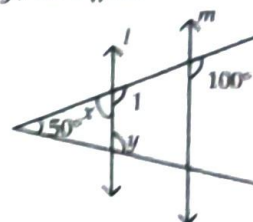
$$\Rightarrow x + x + x = 180^\circ$$

$$\Rightarrow 3x = 180^\circ \Rightarrow x = \frac{180^\circ}{3} = 60^\circ$$

$$\text{Hence, } x = y = 60^\circ$$



18. In the following figure, find the unknown angles x and y , if $l \parallel m$.



Soln.: We have $l \parallel m$

$$\therefore \angle 1 = 100^\circ \quad [\text{Corresponding angles}]$$

$$\text{Now, } \angle 1 + \angle x = 180^\circ \quad [\text{Linear pair}]$$

$$\Rightarrow \angle x = 180^\circ - 100^\circ = 80^\circ$$

$$\text{Now, } \angle y = 50^\circ + \angle x \quad [\text{Exterior angle property}]$$

$$\Rightarrow \angle y = 50^\circ + 80^\circ = 130^\circ$$

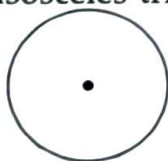
$$\therefore \angle x = 80^\circ \text{ and } \angle y = 130^\circ$$

NCERT Section

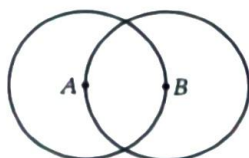
Figure it Out

(Page 150 & 151)

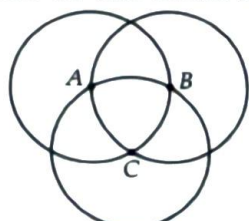
- Use the points on the circle and/or the centre to form isosceles triangles.



- Use the points on the circles and/or their centres to form isosceles and equilateral triangles. The circles are of the same size.



A and B are the centres of circles of the same size



A, B and C are the centres of circles of the same size

Figure it Out

(Page 154)

- We checked by construction that there are no triangles having sidelengths 3 cm, 4 cm, and 8 cm; and 2 cm, 3 cm, and 6 cm. Check if you could have found this without trying to construct the triangle.
- Can we say anything about the existence of a triangle for each of the following sets of lengths?

(a) 10 km, 10 km, and 25 km

(b) 5 mm, 10 mm and 20 mm

(c) 12 cm, 20 cm and 40 cm

You would have realised that using a rough figure and comparing the direct path lengths with their corresponding roundabout path lengths is the same as comparing each length with the sum of the other two lengths. There are three such comparisons to be made.

Figure it Out

(Page 156)

1. Which of the following lengths can be the sidelengths of a triangle? Explain your answers. Note that for each set, the three lengths have the same unit of measure.

- (a) 2, 2, 5 (b) 3, 4, 6
(c) 2, 4, 8 (d) 5, 5, 8
(e) 10, 20, 25 (f) 10, 20, 35
(g) 24, 26, 28

Figure it Out

(Page 159)

1. Check if a triangle exists for each of the following set of lengths:
(a) 1, 100, 100 (b) 3, 6, 9
(c) 1, 1, 5 (d) 5, 10, 12
2. Does there exist an equilateral triangle with sides 50, 50, 50? In general, does there exist an equilateral triangle of any sidelength? Justify your answer.
3. For each of the following, give at least 5 possible values for the third length so there exists a triangle having these as sidelengths (decimal values could also be chosen):
(a) 1, 100 (b) 5, 5
(c) 3, 7

Figure it Out

(Page 161)

1. Construct triangles for the following measurements where the angle is included between the sides:
(a) 3 cm, 75° , 7 cm
(b) 6 cm, 25° , 3 cm
(c) 3 cm, 120° , 8 cm

Figure it Out

(Page 162)

1. Construct triangles for the following measurements:
(a) 75° , 5 cm, 75°
(b) 25° , 3 cm, 60°
(c) 120° , 6 cm, 30°

Figure it Out

(Page 163)

1. For each of the following angles, find another angle for which a triangle is (a) possible, (b) not possible. Find at least two different angles for each category:
(a) 30° (b) 70°
(c) 54° (d) 144°

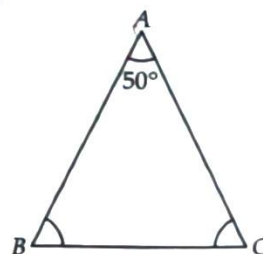
2. Determine which of the following pairs can be the angles of a triangle and which cannot:

- (a) 35° , 150° (b) 70° , 30°
(c) 90° , 85° (d) 50° , 150°

Figure it Out

(Page 165)

1. Find the third angle of a triangle (using a parallel line) when two of the angles are:
(a) 36° , 72° (b) 150° , 15°
(c) 90° , 30° (d) 75° , 45°
2. Can you construct a triangle all of whose angles are equal to 70° ? If two of the angles are 70° what would the third angle be? If all the angles in a triangle have to be equal, then what must its measure be? Explore and find out.
3. Here is a triangle in which we know $\angle B = \angle C$ and $\angle A = 50^\circ$. Can you find $\angle B$ and $\angle C$?

**Figure it Out**

(Page 170 & 171)

1. Construct a triangle ABC with $BC = 5$ cm, $AB = 6$ cm, $CA = 5$ cm. Construct an altitude from A to BC .
2. Construct a triangle TRY with $RY = 4$ cm, $TR = 7$ cm, $\angle R = 140^\circ$. Construct an altitude from T to RY .
3. Construct a right-angled triangle $\triangle ABC$ with $\angle B = 90^\circ$, $AC = 5$ cm. How many different triangles exist with these measurements? [Hint: Note that the other measurements can take any values. Take AC as the base. What values can $\angle A$ and $\angle C$ take so that the other angle is 90° ?]
4. Through construction, explore if it is possible to construct an equilateral triangle that is (i) right-angled (ii) obtuse-angled. Also construct an isosceles triangle that is (i) right-angled (ii) obtuse-angled.

Exercise

Multiple Choice Questions

LEVEL - 1

Triangle Inequality and Angle Sum Property of Triangle

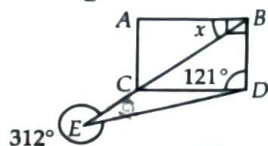
1. Which of the following is not a set of length of the sides of a triangle?

- (a) 8 cm, 4 cm, 20 cm
- (b) 9 cm, 17 cm, 25 cm
- (c) 11 cm, 16 cm, 26 cm
- (d) None of these

(NCERT Page 153) **Ev**

2. Find x in the given figure.

- (a) 68°
- (b) 69°
- (c) 76°
- (d) 79°



(Page 166) **Ev**

3. In a $\triangle ABC$, if $\angle A = \angle B + \angle C$, then find $\angle A$.

- (a) 60°
- (b) 45°
- (c) 90°
- (d) 120°

(Page 166) **U**

4. In an isosceles triangle one of its equal angle is 25° , then the greatest angle is

- (a) 60°
- (b) 95°
- (c) 130°
- (d) 67.5°

(Page 166)

5. $\angle R$, $\angle S$ and $\angle T$ of $\triangle RST$ have measures x , 130° and 25° respectively. What is the value of x ?

- (a) 30°
- (b) 25°
- (c) 55°
- (d) 60°

(Page 166) **Ev**

6. If the angles of triangle are $3x$, $5x$ and $4x$, then find the smallest angle of the triangle.

- (a) 15°
- (b) 45°
- (c) 60°
- (d) 30°

(Page 166)

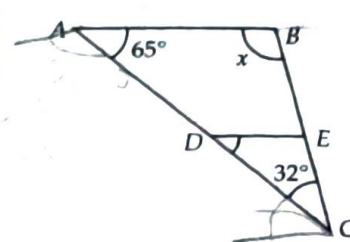
7. In a right isosceles triangle, the measure of one of the angles other than the right angle is

- (a) 90°
- (b) 60°
- (c) 45°
- (d) can't be determined

(Page 166)

8. Find the value of x in the following figure.

- (a) 97°
- (b) 83°
- (c) 115°
- (d) 87°



(Page 166)

9. Which of the following conditions does not guarantee a unique triangle?

- (a) Three sides are given
- (b) Three angles are given
- (c) Two sides and included angle are given
- (d) Two angles and included side is given

An

10. In a triangle, one angle is of 90° . Then from the given statements which is/are true?

- (i) The other two angles can be 45° each.
- (ii) From remaining two angles, one angle is 90° and other is 45° .
- (a) (i) only
- (b) (ii) only
- (c) both (i) and (ii)
- (d) None of these.

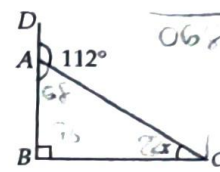
11. Which of the following set of angles cannot be the angles of a triangle?

- (a) 67° , 51° , 62°
- (b) 70° , 83° , 27°
- (c) 90° , 70° , 20°
- (d) 40° , 132° , 18°

Exterior Angles

12. Find x in the given figure.

- (a) 68°
- (b) 122°
- (c) 22°
- (d) 90°



(Page 167)

13. In the given figure, what is the value of $a + b + c$?

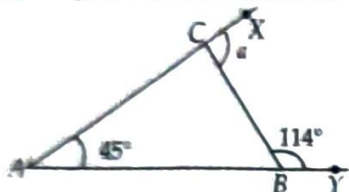
- (a) 180°
(b) 240°
(c) 270°
(d) 360°



(Page 167) **Ap**

14. The side AB of a triangle is produced to Y and side AC is produced to X. $\angle CAB = 45^\circ$ and $\angle CBY = 114^\circ$. Find value of a .

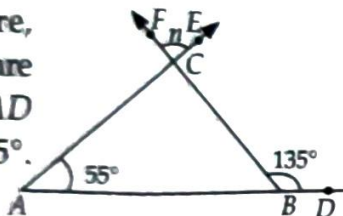
- (a) 68°
(b) 45°
(c) 111°
(d) 57°



(Page 167)

15. In the given figure, ABD, ACE and BCF are straight lines $\angle EAD = 55^\circ$, $\angle CBD = 135^\circ$. Find value of n .

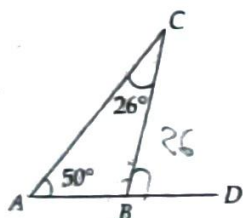
- (a) 55°
(b) 67.5°
(c) 80°
(d) 100°



(Page 167) **Ev**

16. In $\triangle ABC$, find $\angle DBC$.

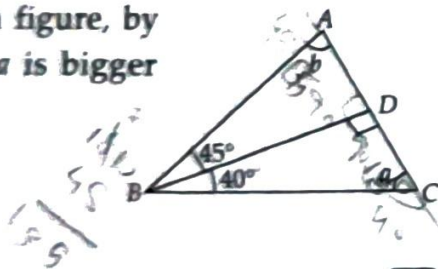
- (a) 45°
(b) 50°
(c) 76°
(d) 104°



(Page 167)

17. In the given figure, by how much a is bigger than b ?

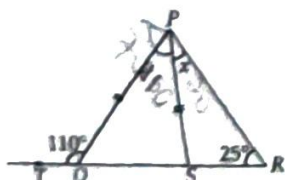
- (a) 50°
(b) 45°
(c) 5°
(d) 10°



(Page 167) **Ap**

18. In the given figure, $PQ = PS$. The value of x is

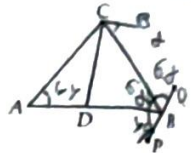
- (a) 35°
(b) 45°
(c) 55°
(d) 70°



(Page 167)

19. In the given figure below, AC is parallel to PQ, $CD = CB$, $\angle BAC = 44^\circ$ and $\angle CBQ = 68^\circ$. Find $\angle ACD$.

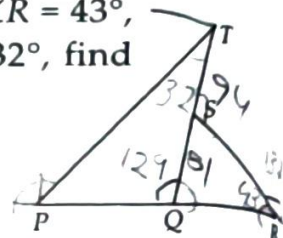
- (a) 24°
(b) 44°
(c) 32°
(d) 28°



(Page 166, 167) **Ev**

20. In the given figure, if $\angle R = 43^\circ$, $\angle RST = 94^\circ$ and $\angle T = 32^\circ$, find $\angle P$.

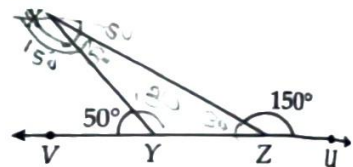
- (a) 21°
(b) 62°
(c) 51°
(d) 19°



(Page 167)

21. In the given figure, the two exterior angles are given. Find $\angle YXZ$.

- (a) 20°
(b) 50°
(c) 30°
(d) 40°



22. The exterior angle of a triangle is 75° . The pair of interior opposite angles can be

- (a) $100^\circ, 25^\circ$
(b) $40^\circ, 35^\circ$
(c) $40^\circ, 50^\circ$
(d) $80^\circ, 10^\circ$

An

Altitudes of a Triangle

23. The altitude of a triangle is always drawn

- (a) from a vertex to the opposite vertex.
(b) from a vertex perpendicular to the opposite side.
(c) from the midpoint of a side to the opposite vertex.
(d) from a side to the midpoint of the opposite side.

(Page 168) **U**

24. Which type of triangle can have a side that is also an altitude?

- (a) Equilateral triangle
(b) Isosceles triangle
(c) Right triangle
(d) All of these

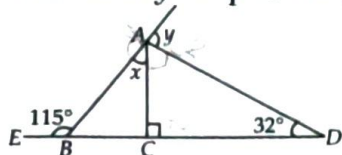
(Page 169) **An**

25. The altitude of the obtuse angled triangle lies

- (a) inside the triangle
- (b) outside the triangle
- (c) Both (a) and (b)
- (d) coinciding with a side

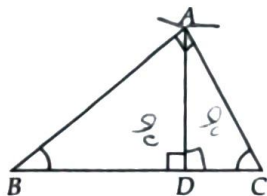
An

26. The value of x and y respectively are



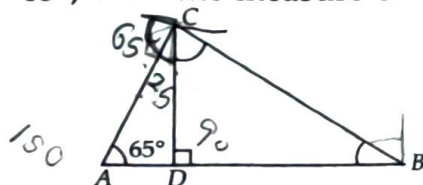
- (a) $58^\circ, 72^\circ$
- (b) $32^\circ, 58^\circ$
- (c) $72^\circ, 58^\circ$
- (d) $25^\circ, 97^\circ$

27. In $\triangle ABC$, $\angle BAC = 90^\circ$ and $AD \perp BC$. If $\angle BAD = 40^\circ$, then what is the measure of $\angle ACD$?



- (a) 45°
- (b) 30°
- (c) 50°
- (d) 40°

28. $\triangle ABC$ is right angled at C and $CD \perp AB$. If $\angle A = 65^\circ$, then the measure of $\angle BCD$ is



- (a) 25°
- (b) 75°
- (c) 65°
- (d) None of these

Ev

29. Which of the following statements is incorrect?

- (a) The sum of any two sides of a triangle is greater than the third side.
- (b) A triangle can have all its angle acute.
- (c) A right-angled triangle cannot be equilateral.
- (d) Difference of any two sides of a triangle is greater than the third side.

U

30. If two angles of a triangle measure 63° and 27° , then it is a/an

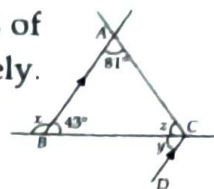
- (a) Acute-angled triangle
- (b) Obtuse-angled triangle
- (c) Right-angled triangle
- (d) None of these

An

LEVEL - 2

31. In the figure, find the values of the angles x , y and z respectively.

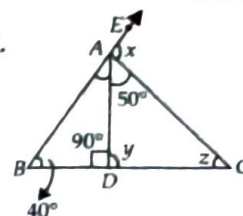
- (a) $137^\circ, 43^\circ, 56^\circ$
- (b) $122^\circ, 56^\circ, 48^\circ$
- (c) $116^\circ, 43^\circ, 56^\circ$
- (d) $137^\circ, 56^\circ, 48^\circ$



Ev

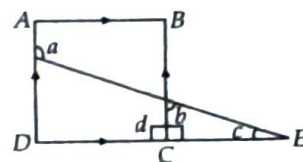
32. Find the value of $x + y + z$.

- (a) 180°
- (b) 225°
- (c) 195°
- (d) 210°

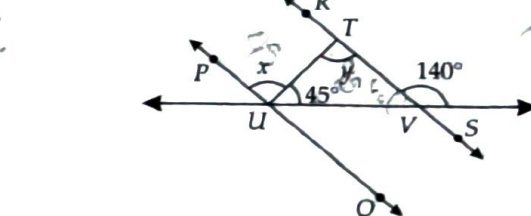


33. In the figure, if $\angle d$ is a right angle, then $\angle a + \angle b =$

- (a) 180°
- (b) 90°
- (c) 75°
- (d) 60°

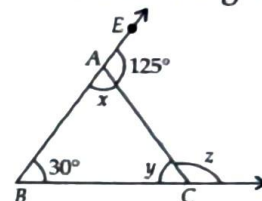


34. In the given figure, $PQ \parallel RS$. Find the values of x and y respectively.



- (a) $85^\circ, 85^\circ$
- (b) $95^\circ, 75^\circ$
- (c) $85^\circ, 95^\circ$
- (d) $95^\circ, 95^\circ$

35. Find the value of z in the given figure.



- (a) 95°
- (b) 85°
- (c) 55°
- (d) 75°

36. In a $\triangle DEF$, $\angle E = 70^\circ$ and $\angle F = 60^\circ$. The bisector of $\angle F$ meets DE in G . The values of $\angle DGF$ and $\angle FGE$ respectively are

- (a) $105^\circ, 75^\circ$
- (b) $130^\circ, 50^\circ$
- (c) $100^\circ, 80^\circ$
- (d) $90^\circ, 90^\circ$

U

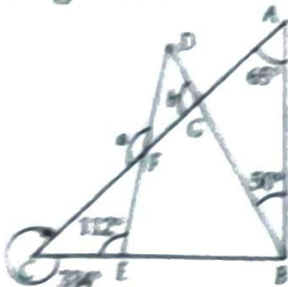
37. $\angle A$ and $\angle B$ are the interior opposite angles of $\angle BCD$ in $\triangle ABC$. Which of these statements is true?



- (a) $\angle BCA = \angle BCD - \angle A$
 (b) $\angle A + 180^\circ = \angle B$
 (c) $\angle A = 90^\circ - \angle B$
 (d) $\angle B = \angle BCD - \angle A$

An

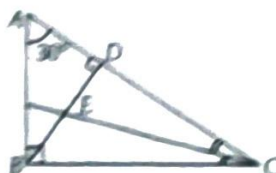
38. In the given figure, the value of $a - b$ is



- (a) 65° (b) 138° (c) 73° (d) 48°

39. In the given figure, $AB \perp BC$, $BD \perp AC$, $\angle A = 30^\circ$ and CE bisects $\angle C$. Find $\angle CED$.

- (a) 30°
 (b) 60°
 (c) 45°
 (d) 65°



40. If measure of one side of an isosceles triangle is 5 cm then what could be the measure of other two sides?

- (a) 5 cm, 6 cm (b) 3 cm, 3 cm
 (c) 5 cm, 7 cm (d) All of these

41. In the given figure $AB + BC + CD + DA$ is less than

- (a) $AC + BD$
 (b) $2(AC + BD)$
 (c) $AD + BC$
 (d) $AB + 2AC$



42. In which of the following triangle the three altitude do not intersect at a point?

- (a) Obtused angled triangle
 (b) Acute angled triangle
 (c) Right angled triangle
 (d) None of these

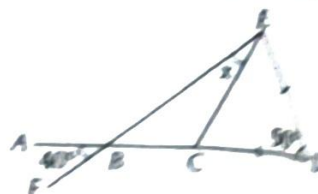
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LEVEL - 3 (HOTS)

Competency Focused Questions (CFQs)

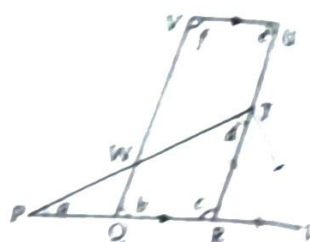
43. In the given figure, ACD is a straight line. Then value of x is

- (a) 25°
 (b) 35°
 (c) 45°
 (d) 55°



44. The given figure is not drawn to scale. Which one of the following options is true?

- (a) $\angle a + \angle d = 60^\circ$
 (b) $\angle c + \angle e = 160^\circ$
 (c) $\angle c + \angle f = 240^\circ$
 (d) $\angle e + \angle f = 180^\circ$



An

45. Which of the following is correct relation between x and y ?

- (a) $x > y$
 (b) $x < y$
 (c) $x = y$
 (d) $\frac{x}{y} = \frac{1}{2}$



Ap

46. In a triangle, there are 3 different angles. Which of the following statements is/are definitely true about the angles?

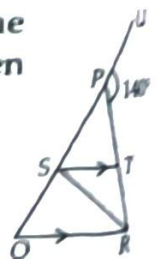
- (i) Sum of all the angles is 100° .
 (ii) One angle is obtuse and other two are acute and right angle respectively.
 (iii) Sum of any two angles is always less than 180° .

- (a) (i) only
 (b) (ii) and (iii) both
 (c) (iii) only
 (d) (i) and (iii) both

U

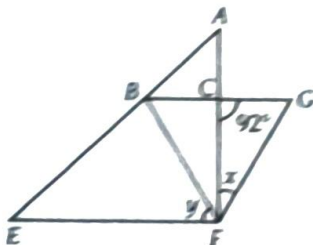
47. In the given figure, $PQ = PR$, $RS = RQ$ and $ST \parallel QR$. If the exterior angle RPU is 140° , then the measure of angle TSR is

- (a) 55°
 (b) 40°
 (c) 50°
 (d) 45°



48. In $\triangle ABC$, $\angle A = 100^\circ$, AD bisects $\angle A$ and $AD \perp BC$. Then, $\angle B$ is equal to
(a) 80° (b) 20° (c) 40° (d) 30°

49. In the given figure (not drawn to scale), if EFA is a right-angled triangle with $\angle EFA = 90^\circ$ and FGB is an equilateral triangle, then the value of $y - 2x$ is



- (a) 2° (b) 8°
(c) 17° (d) 20°

Ev

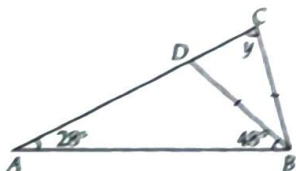
50. A triangle can be constructed by taking its sides as _____

- (a) 1.8 cm, 2.6 cm, 4.4 cm
(b) 2 cm, 3 cm, 4 cm
(c) 2.4 cm, 2.4 cm, 6.4 cm
(d) 3.2 cm, 2.3 cm, 5.5 cm

Cr

51. In the given figure (not drawn to scale), ABC is a triangle and $BC = BD$. Find the value of y .

- (a) 73°
(b) 74°
(c) 75°
(d) 76°



52. Read the statements carefully and state 'T' for true and 'F' for false.

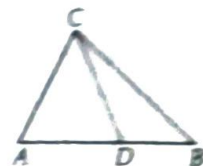
- (i) The difference between the lengths of any two sides of a triangle is larger than the length of the third side.
(ii) The measure of any exterior angle of a triangle is equal to the sum of the measures of its interior opposite angles.
(iii) In an equilateral triangle, only one angle is acute.
(iv) The line segment joining a vertex of a triangle to its opposite side is called an altitude of the triangle.

(i) (ii) (iii) (iv)

- (a) T F T F
(b) F T F F
(c) F F T T
(d) F T T F

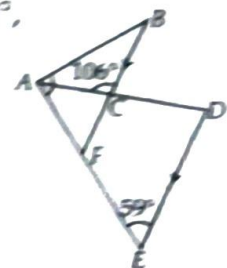
53. In the given figure, if $\angle A = 2\angle B$ and $\angle ACD = 2\angle DCB$, then find the measure of $\angle DCB + \angle B$.

- (a) 60°
(b) 90°
(c) 45°
(d) 75°



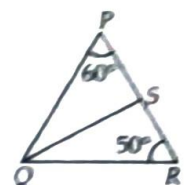
54. In the given figure, ACD is a straight line. $\angle ACB = 106^\circ$, $AC = CB$ and $BF \parallel DE$. Find $\angle BAE$.

- (a) 42°
(b) 84°
(c) 39°
(d) None of these



55. In the given figure (not drawn to scale), QS is the bisector of $\angle PQR$. Find the value of $2\angle QSP$.

- (a) 85°
(b) 70°
(c) 170°
(d) 140°



56. Which of the following statements is CORRECT?

- (a) If lengths of any two sides of a triangle are 7 cm and 10 cm, then the length of its third side lies between 3 cm and 17 cm.
(b) It is not possible to construct a unique triangle, if all its three sides are given.
(c) An angle of 75° can't be drawn using protractor and ruler.
(d) None of these

An

57. Arrange the given steps in CORRECT order, while constructing $\triangle PQR$ where $QR = 4.2$ cm, $\angle Q = 120^\circ$, $PQ = 3.5$ cm and $PM \perp RQ$ (produced).

Step 1 : Now, extend RQ to S and with P as centre and with a sufficient radius, draw an arc, cutting SR at A and B .

Step 2 : Along QX , cut off $QP = 3.5$ cm.

Step 3 : Draw a line segment $QR = 4.2$ cm and construct $\angle RQX = 120^\circ$.

Step 4 : Joint PR .

Step 5 : Joint PC , meeting RQ produced at M . Then, $PM \perp RQ$ (produced).

Step 6 : With A and B as centres and radius more than half AB , draw two arcs cutting each other at C .

(a) $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6$

(b) $3 \rightarrow 2 \rightarrow 4 \rightarrow 6 \rightarrow 5 \rightarrow 1$

(c) $2 \rightarrow 4 \rightarrow 3 \rightarrow 1 \rightarrow 5 \rightarrow 6$

(d) $3 \rightarrow 2 \rightarrow 4 \rightarrow 1 \rightarrow 6 \rightarrow 5$



58. Which of the following steps is INCORRECT while constructing $\triangle XYZ$ if it is given that $XY = 6$ cm, $\angle ZXY = 30^\circ$ and $\angle XYZ = 100^\circ$

Step 1 : Draw line XY of length 6 cm.

Step 2 : At X , draw a ray XP making an angle of 30° with XY .

Step 3 : At Y , draw a ray YQ making an angle of 100° with YX .

Step 4 : The point of intersection of the two rays XY and YQ is Z .

(a) Step 1 (b) Step 2 and Step 4

(c) Step 3 (d) Step 4



59. You wanted to make a triangular frame for the modern art painted by your mother. She says that two sides of the frame should be 7 dm and 14 dm and the angle formed between them is 45° . What could be the ideal length for the third side of the triangular frame?

(a) 21 dm (b) 6 dm
(c) 10 dm (d) None of these

60. In which type of triangle can all three altitude divides the opposite side into two equal parts?

(a) Right angle triangle
(b) Equilateral triangle
(c) Isosceles triangle
(d) (b) and (c) both



Match the Following

In this section each question has two matching lists. Choices for the correct combination of elements from List-I and List-II are given as options (a), (b), (c) and (d) out of which, one is correct.

1. Match the following.

List-I

List-II

- (P) In a $\triangle ABC$, if $\angle A = 30^\circ$, $\angle B = 50^\circ$, then $\angle C = ?$ 1. 130°
(Q) In a $\triangle ABC$, BC is extended to D . If $\angle ABC = 90^\circ$, $\angle BAC = 40^\circ$, then $\angle ACD = ?$ 2. 90°
(R) In a right angled triangle, measure of greatest angles is 3. 100°
(S) In a $\triangle ABC$, if $\angle A = x$, $\angle B = 2\angle A$, $\angle C = 3\angle A$, then $x =$ 4. 30°
(a) P-3, Q-2, R-4, S-1
(b) P-2, Q-3, R-1, S-4
(c) P-3, Q-1, R-2, S-4
(d) P-2, Q-4, R-3, S-1

2. Match the following.

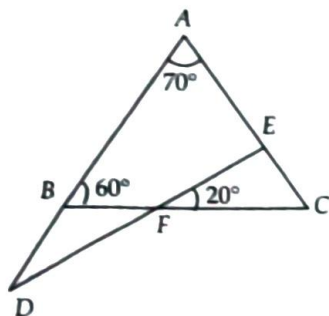
List-I

List-II

- (P) In an isosceles right angled triangle whose longest side measure 10 cm, the other side has measure 1. 3 cm
(Q) Measure of two sides of a triangle is 4 cm and 2 cm then measure of third side can be (If the third side has longest measure) 2. 4 cm
(R) Measure of one side of an equilateral triangle is 2 cm then sum of measure of other two sides will be 3. 5 cm
(S) Measure of two equal sides of an isosceles triangle is 2 cm each then measure of third side can be 4. 6 cm

- (a) P-2, Q-1, R-3, S-4
 (b) P-4, Q-3, R-2, S-1
 (c) P-4, Q-2, R-3, S-1
 (d) P-3, Q-2, R-1, S-2

3. On the basis of given figure, match the following.



List-I

- (P) Measure of $\angle DBC$ is
 (Q) $\angle BFD$ is equal to
 (R) $\angle BDF$ has measure
 (S) $\angle CEF$ is equal to
 (a) P-3, Q-1, R-4, S-2
 (b) P-2, Q-3, R-4, S-1
 (c) P-2, Q-4, R-3, S-1
 (d) P-3, Q-4, R-1, S-2

List-II

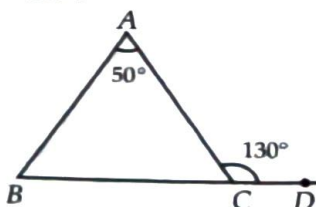
1. 20°
 2. 110°
 3. 120°
 4. 40°

Assertion & Reason Type

Directions : In each of the following questions, a statement of Assertion is given followed by a corresponding statement of Reason just below it. Of the statements, mark the correct answer as

- (a) If both assertion and reason are true and reason is the correct explanation of assertion.
 (b) If both assertion and reason are true but reason is not the correct explanation of assertion.
 (c) If assertion is true but reason is false.
 (d) If assertion is false but reason is true.

1. **Assertion :** In $\triangle ABC$, measure of $\angle ABC$ is 80° .

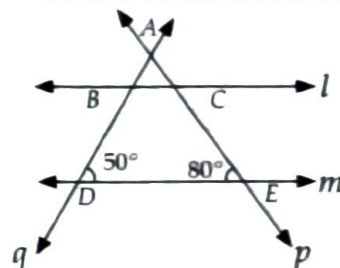


Reason : If a side of a triangle is produced then the exterior angle so formed is equal to the sum of the two interior opposite angles.

2. **Assertion :** The triangle is possible to be formed by three line segments of lengths 3 cm, 4 cm, 5 cm.

Reason : The sum of any two sides of a triangle is less than the third side.

3. **Assertion :** In the given figure, if $l \parallel m$, then $\triangle ABC$ is an isosceles triangle.



Reason : If all sides of a triangle are equal, then it is an isosceles triangle.

4. **Assertion :** For a triangle, largest side is taken as base.

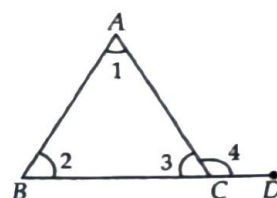
Reason : This ensures the triangle fits properly within drawing surface.

5. **Assertion :** The altitude of a right angled triangle can not be drawn using a set square and ruler.

Reason : The perpendicular sides of a right angled triangle are itself altitudes of triangle.

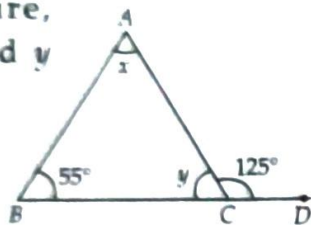
Comprehension Type

PASSAGE-I : If a side of a triangle is produced then the exterior angle so formed is equal to the sum of the two interior opposite angles.
 i.e., $\angle 1 + \angle 2 = \angle 4$



1. In the given figure, the values of x and y respectively are

- (a) $75^\circ, 50^\circ$
 (b) $95^\circ, 45^\circ$
 (c) $70^\circ, 55^\circ$
 (d) $90^\circ, 45^\circ$



2. One side of a triangle is produced and the exterior angle so formed is 120° . If the interior opposite angles are in the ratio 3 : 5, find the measure of each angle of the triangle.

- (a) $45^\circ, 75^\circ, 60^\circ$ (b) $40^\circ, 70^\circ, 70^\circ$
 (c) $45^\circ, 70^\circ, 65^\circ$ (d) $40^\circ, 75^\circ, 65^\circ$

3. If the sides of a triangle are produced in order, then the sum of the exterior angles so formed is

- (a) 180° (b) 270° (c) 90° (d) 360°

PASSAGE-II : A group of students were given straws of different lengths to test whether they could form triangles. The straws had length 4 cm, 5 cm, 6 cm, 10 cm and 15 cm. The teacher reminded them to use triangle inequality.

1. Which of the following sets of straws can form a triangle?

- (a) 4 cm, 6 cm, 10 cm
 (b) 5 cm, 6 cm, 10 cm
 (c) 5 cm, 10 cm, 4 cm
 (d) 4 cm, 15 cm, 6 cm

2. If a student has chosen two straws of length 6 cm and 15 cm, what should be length of third straw to form a triangle?

- (a) 4 cm (b) 5 cm
 (c) 10 cm (d) None of these

3. A student claim that "If the longest straw is greater than or equal to the sum of the other two, no triangle can be formed". Which of the following examples support the given claim?

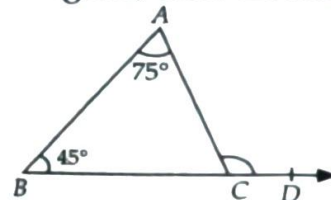
- (a) 4 cm, 5 cm, 6 cm
 (b) 4 cm, 10 cm, 5 cm
 (c) 5 cm, 6 cm, 10 cm
 (d) 6 cm, 10 cm, 15 cm

Subjective Problems

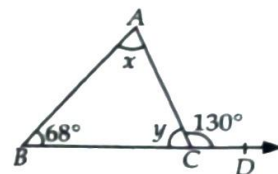
Very Short Answer Type

1. In a $\triangle ABC$, $\angle A = 65^\circ$, $\angle B = 35^\circ$, find the measure of $\angle C$.

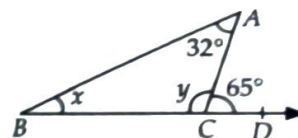
2. In the given figure, find $\angle ACD$.



3. In the figure given, find the value of x and y .



4. In the given figure, find the value of x and y .



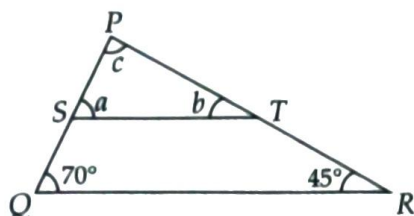
5. Is it possible to draw a triangle whose sides are 5 cm, 8 cm, 9 cm?
6. What is the measure of each angle of an equilateral triangle?
7. In a $\triangle XYZ$, if $\angle X = 90^\circ$ and $\angle Z = 48^\circ$, find $\angle Y$.
8. In $\triangle ABC$, Find $\angle A + \angle B$, if the exterior angle of $\angle C$ is 135° .
9. How many measures are needed to draw a triangle?
10. Can we draw altitude of a triangle without ruler?

Short Answer Type

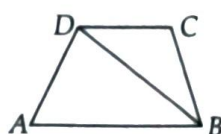
1. Find the angles of a $\triangle ABC$ which are in the ratio 11 : 13 : 12.
2. Is it possible to draw a triangle whose sides are 6 cm, 8 cm and 14 cm?

3. Two angles of triangle are equal and the third angle measures 80° . Find the measure of each of the unknown angles.

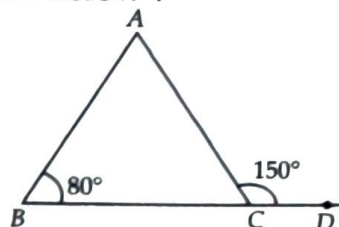
4. In $\triangle PQR$, $QR \parallel ST$, find the value of a , b and c .



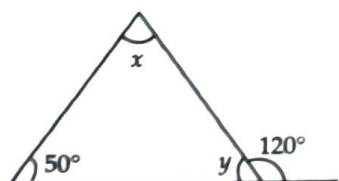
5. The given figure is formed up of two triangles. Find $\angle ABD + \angle DBC + \angle BCD + \angle CDB + \angle BDA + \angle DAB$.



6. Find all the angles of the triangle in the figure given below :



7. Construct a right angled $\triangle PQR$, where $\angle Q = 90^\circ$, $QR = 8$ cm and $PR = 10$ cm.
8. Find the value of x and y in the given diagram.

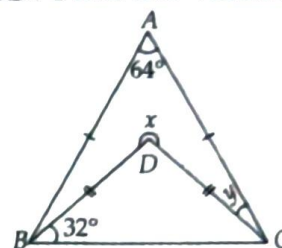


9. Name types of triangles based on sides and angles.
10. Construct $\triangle ABC$ with $AB = 5$ cm, $BC = 8$ cm and $\angle ABC = 60^\circ$.

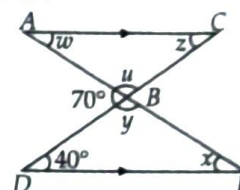
Long Answer Type

1. Construct $\triangle PQR$, if $PQ = 5$ cm, $\angle PQR = 105^\circ$ and $\angle QRP = 40^\circ$.
2. Examine whether you can construct $\triangle DEF$ such that $EF = 7.2$ cm, $\angle E = 110^\circ$ and $\angle F = 80^\circ$. Justify your answer.

3. In the given figure, $\angle DBC = 32^\circ$, $AB = AC$ and $BD = CD$. Find the values of x and y .



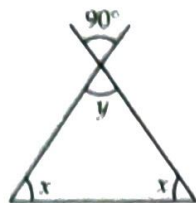
4. In the figure given below, find the angles x , y , z , u and w .



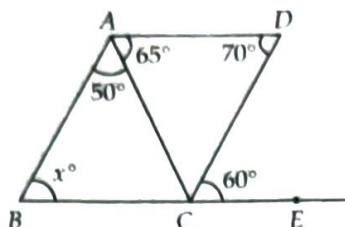
5. In a $\triangle PQR$, if $6\angle R = 4\angle Q = 3\angle P$, then find $\angle P$, $\angle Q$, $\angle R$.

Integer/Numerical Value Type

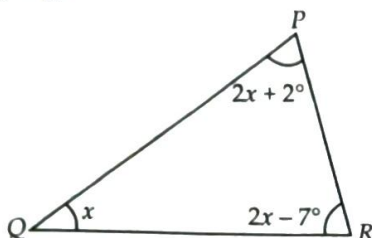
1. If an exterior angle of a triangle is 145° and one of the interior opposite angles is 75° , then other angle is $14^\circ \times K$. Find K .
2. The lengths of two sides of a triangle are 5 cm and 7 cm. If the length of the third side fall between K and 12, then find K .
3. Two sides of a triangle are 13 cm and 20 cm. If the third side has an integer measure then how many such triangle are possible?
4. In a triangle, one angle is twice the second and third angle is three times the second angle. Find the value of largest angle (in degrees).
5. In an isosceles triangle the measure of longest side is 12 cm. What should be the length of other two sides so that the largest triangle [Measure of sides integer] can be made?
6. The angles of a triangle are in the ratio $6 : 5 : 4$. Find the unit digit of the largest angle.
7. Find the value of $x + y$ (in degree) from the given figure.



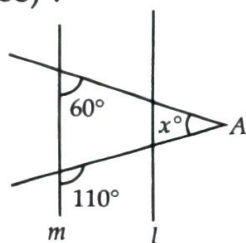
8. Find the value of x (in degree) in the given figure.



9. Find the value of x (in degree) in the adjoining figure.

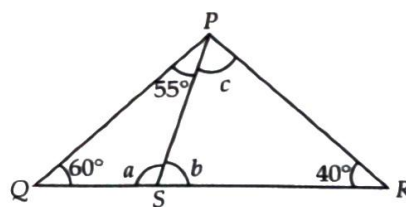


10. In the given figure $l \parallel m$, then find the value of x (in degree).



- What will be measure of $\angle A$ in $\triangle ABC$?
(a) 60° (b) 75°
(c) 85° (d) 90°
- What would be measure of $\angle BAD$?
(a) 60° (b) 50°
(c) 30° (d) 90°
- Why is the base BC drawn first in the construction?
(a) It is the longest side.
(b) Angles can not be drawn without the base line.
(c) Drawing line is easy.
(d) None of these
- After constructing $\triangle ABC$, which instrument is most suitable for drawing altitude?
(a) Ruler only
(b) Protractor and ruler
(c) Set square and ruler
(d) Divider and ruler
- What would be the measure of altitude from A to BC ?
(a) 5 cm (b) 6 cm
(c) 7 cm (d) None of these

Case II : Aayush's sister made a triangular photo frame as a part of her art and craft project given in her school. By looking at the triangular shape of the frame, he recalled, how his mother helped him in solving questions based on the given figure. Aayush has listed below the questions to solve them again. Help Aayush to achieve the correct answers.



- The value of a is
(a) 55° (b) 65° (c) 45° (d) 75°
- The value of b is
(a) 115° (b) 105° (c) 95° (d) 65°

Case Based Questions

Case I : A city planner is designing a triangular park. The park must follow some geometric rules so that walking paths can be constructed.

The planner made a design on sheet in which, for length in metres, he used cm on sheet to get the accurate measurements.

The base (BC) of triangular park is 6 m long, angle constructed at B was 60° and at C was 45° . The lines drawn meet each other at point A so that $\triangle ABC$ is formed. Later, the planner wants to add an altitude (AD) from A to BC to design a path.

Based on the above information, answer the following questions.

CFQs

3. The value of c can be found by which of the following ways?

- (a) $c = 180^\circ - 40^\circ - 115^\circ$
- (b) $c = 180^\circ - 40^\circ - 60^\circ - 55^\circ$
- (c) $c = 65^\circ - 40^\circ$
- (d) All of these

4. The longest side in ΔPQR is

- (a) QR (b) PR
- (c) PQ (d) can't be determined

5. The shortest side in ΔPRS is

- (a) PR
- (b) PS
- (c) RS
- (d) can't be determined

Case III : Raghav is preparing for his forthcoming assignment on triangles. For which, he is solving the worksheet given by his teacher and comes across the following questions. He finds it difficult to answer



and so he decides to get them clear by his teacher in the school by the next day. Answer these questions to test your understanding.

CFQs

1. Which of the following statements is incorrect?

- (a) No triangle can have its altitude outside the triangle.
- (b) Altitude of a triangle is drawn from vertex to its opposite side.
- (c) In an isosceles triangle, the altitude drawn from the vertex divides triangle into two parts of equal area.
- (d) All of these

2. How many altitudes does a right-angled triangle have?

- (a) 1 (b) 2
- (c) 3 (d) can't be determined

3. Which of the following statements is correct?

- (a) It is possible to have a triangle in which two of the angles are obtuse.
- (b) It is possible to have a triangle in which two angles are acute.

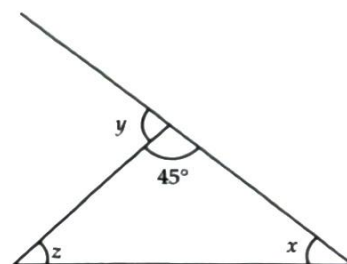
(c) It is possible to have a triangle in which each angle is less than 60° .

(d) It is possible to have a triangle in which each angle is greater than 60° .

4. Which of the following is possible in a ΔPQR ?

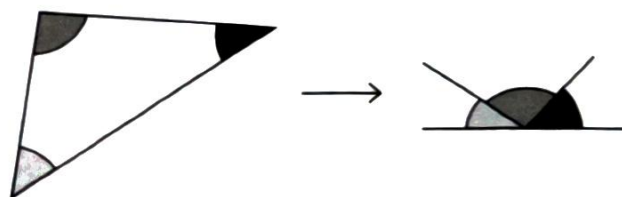
- (a) $PQ - QR > PR$
- (b) $PQ + QR < PR$
- (c) $PQ - QR < PR$
- (d) $PQ + PR < QR$

5. In the adjoining figure, if x is two-third of z , then the value of z is

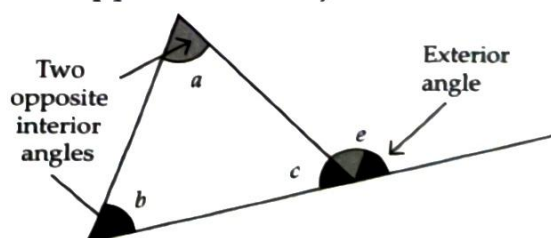


- (a) 225° (b) 54° (c) 81° (d) 99°

Case IV : In a class, the teacher explained the angle sum property of a triangle practically by cutting a piece of triangle into three parts such that each part has a vertex. Then by placing all the vertices together in such a way that they form a straight angle.



She also extended her practical teaching. She also explained the concept of exterior angle property of a triangle, by showing how an exterior angle of a triangle is equal to the sum of its two opposite non-adjacent interior angles.



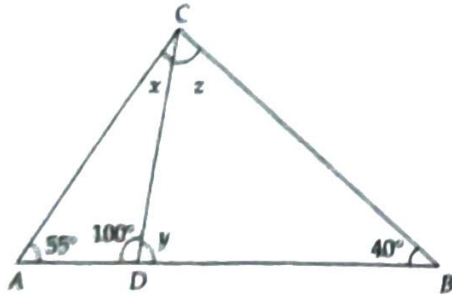
Based on the above information, answer the following questions.

CFQs

1. In a triangle, if two angles sum up to 90° , then the triangle is a

- (a) isosceles triangle
- (b) scalene triangle
- (c) right angled triangle
- (d) can't be determined

2. The value of $x + z$ in the given figure is



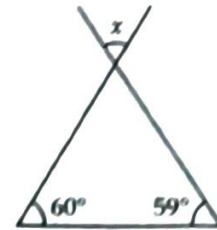
(a) 75°

(b) 85°

(c) 80°

(d) 65°

3. In the given figure, the measure of the angle x is



(a) 119°

(b) 19°

(c) 61°

(d) 111°

4. In $\triangle ABC$, If $AB = CB$ and $\angle C = 40^\circ$, then the measure of exterior angle at B is

(a) 100°

(b) 40°

(c) 50°

(d) 80°